

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 11 · LECTURE NOTES

Association and Confounding

Finding variants, and the structure that fakes them.

MODULE OUTCOMES

- Explain the association study design.
- Explain population structure as a confounder.
- Explain linkage disequilibrium and what it costs you.
- Explain effect sizes and missing heritability.
- State what an association establishes.

1 The design

TEST EVERY VARIANT FOR ASSOCIATION WITH THE TRAIT, WHICH IS A MILLION TESTS AND A CORRELATIONAL DESIGN

Genotype many individuals at a million or more variant positions, and test each position for association with the trait — which is Module 10 §3's multiple testing problem at a considerably larger scale, and is why the field uses a genome-wide significance threshold of 5×10^{-8} .

That threshold is a Bonferroni correction for roughly a million effectively independent tests — derived rather than conventional — which is a real point in the field's favour: it adopted a stringent, justified threshold early, and its replication record improved markedly as a result.

And the design is observational — there is no intervention, so any association found may be confounded by anything correlated with both genotype and trait — which is §2's subject.

Plus genotype is not randomised across the population — it is correlated with ancestry, which is correlated with geography, environment, diet, and access to healthcare — and that is the specific confounder this field had to solve before any of its results could be believed.

2 Population structure

THE MECHANISM

allele frequencies differ between ancestral populations
so does almost every trait, for reasons of environment, diet, access, and history
sample both populations unevenly across cases and controls, and EVERY differentiated variant associates with the trait

THE CLASSIC ILLUSTRATION

a variant common in one ancestry group
"associates" with a trait that is more common in that group, with no causal relationship at all

THE CORRECTIONS

principal components of the genotype matrix,
included as covariates in the model
linear mixed models with a relatedness matrix
and these work well for broad structure and less well for fine-scale or recent structure

WHICH IS WHY ANCESTRY MUST BE REPORTED, always.

The corrections work well for broad structure and less well for fine-scale or recent structure — so residual confounding is expected rather than excluded, and the right statement is 'after correcting for the first ten principal components' rather than 'after correcting for population structure'. **Population structure is the field's canonical confounder, and the corrections are partial** — which is a good example of a known confounder handled well enough to proceed and not well enough to forget.

3 Linkage disequilibrium

NEARBY VARIANTS ARE INHERITED TOGETHER, SO A TESTED VARIANT STANDS IN FOR ITS NEIGHBOURS

Recombination is rare over short distances, so variants near one another on a chromosome are statistically correlated across a population — which is linkage disequilibrium, and it decays with distance and with the number of generations since the variants arose.

Which is exactly why genotyping a million variants can cover a three-billion-base genome — each tested variant tags a block of its neighbours — and that is what makes the whole design affordable rather than requiring full sequencing of everyone.

And it means the associated variant is usually not the causal one — it is merely correlated with it — so fine-mapping to identify the actual causal variant within an associated block is a separate and genuinely difficult problem, often unresolved for years after the association.

Plus the great majority of associated variants are non-coding — regulatory rather than protein-altering — which makes establishing the mechanism substantially harder than establishing the association, since you must identify not only the variant but the gene it regulates and the tissue in which it does so.

4 Effect sizes

- Individual effect sizes are tiny — odds ratios close to one, each explaining a fraction of a percent of trait variance — so no single common variant predicts anything useful about an individual, whatever a direct-to-consumer report may imply.
- Which is why enormous sample sizes are required — hundreds of thousands of participants to detect effects of that size — and why the field consolidated into large international consortia and biobanks, which has its own consequences for who is represented.
- And polygenic scores aggregate many small effects into a single number — predictive at the population level and weakly so for individuals — and they transfer poorly to ancestries not represented in the training cohort, which is a direct consequence of §§2 and 3: different allele frequencies and different linkage patterns mean the tag variants no longer tag the same thing.
- Plus missing heritability: the known variants together explain far less of the variance than twin and family studies suggest is heritable — and rare variants, gene-gene and gene-environment interactions, and problems with the heritability estimates themselves are all live candidate explanations.
- And a heritability estimate is specific to a population and an environment: it is the proportion of observed variance in that setting attributable to genetic variance there. It is not a property of a trait, and reading it as one is the characteristic error — a highly heritable trait in one environment can be barely heritable in another, with no genetics having changed.

SO WHAT AN ASSOCIATION ESTABLISHES

'A variant in this region is statistically associated with this trait, in this cohort, after correcting for this much population structure, at this significance threshold' — which is a real and useful finding and has repeatedly pointed to genuine biology.

And not: that this particular variant is causal (§3's linkage disequilibrium), **that the mechanism is known, or that it predicts anything actionable for an individual person** (§4's effect sizes).

The honest chain is: association → causal variant → affected gene → mechanism → intervention, and each of those arrows is years of work that the association itself does not perform.

Which is Module 01 §3's inference chain at its longest — and **the reason this field's results are so frequently overstated in reporting is structural: the first link is cheap and automatable, and every subsequent one is not.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Visscher et al. — 10 years of GWAS discovery (free)	The whole module — an honest assessment by people in the field, including the overclaiming.
Price et al. — Principal components analysis corrects for stratification (free)	§2's correction , in the original.
Martin et al. — Clinical use of current polygenic risk scores may exacerbate health disparities (free)	§4's transferability problem , measured — and the consequences stated.
Manolio et al. — Finding the missing heritability of complex diseases (free)	§4 — the problem stated, with the candidate explanations laid out.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Compute the Bonferroni threshold** for a million tests and compare to the field's.
- 2 **Simulate two populations** with different allele frequencies and different trait rates.
- 3 **Sample them unevenly** and count the spurious associations.
- 4 **Correct with principal components** and recount.

- 5 **Make the structure fine-scale** and watch the correction weaken.
- 6 **Simulate linkage** and show a non-causal variant associating more strongly than the causal one.
- 7 **Compute the variance explained** by a published effect size.
- 8 **Apply a polygenic score** across two simulated ancestries.
- 9 **Explain heritability** to somebody, correctly, in four sentences.
- 10 **Find a news article** about a gene for a trait, and write the accurate version.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Describe the design and its two structural weaknesses.
- 2 Where does the genome-wide threshold come from?
- 3 Explain how population structure fakes associations.
- 4 Name two corrections, and state their limits.
- 5 What is linkage disequilibrium, and what does it buy?
- 6 What does it cost, and what is fine-mapping?
- 7 Why are most associated variants hard to interpret?
- 8 How large are effect sizes, and what follows?
- 9 Why do polygenic scores transfer poorly?
- 10 State what a heritability estimate is, and the common error.